

EX NO: 4

IT22030 ETHICAL HACKING LABORATORY

DATE:

Wireless Network Hacking and Cloud Computing

AIM: To recover saved Wi-Fi network passwords on a Windows machine using built-in netsh commands, and to create a free-tier virtual machine on a public cloud platform (AWS / GCP) to demonstrate cloud computing.

PROCEDURE:

1. Open Command Prompt as Administrator and execute 'netsh wlan show profiles' to list all saved Wi-Fi network profiles on the system.

```
C:\Windows\System32>netsh wlan show profiles

Profiles on interface Wi-Fi:

Group policy profiles (read only)
-----
    <None>

User profiles
-----
    All User Profile : Asd
    All User Profile : POCO M4 5G
    All User Profile : Loganathan
    All User Profile : Loganathan_5G
    All User Profile : POCO M7 5G
    All User Profile : Anna_WiFi
    All User Profile : Blaze's Mobile
    All User Profile : Recruitment
    All User Profile : Modern Computing Lab
    All User Profile : Guest
    All User Profile : vivo Y93
    All User Profile : Logesh's Hotspot
    All User Profile : Dax's nothing
    All User Profile : med
    All User Profile : wifi
```

2. Pick any saved profile name (for example: Asd) and run 'netsh wlan show profile name="Asd" key=clear'. Under 'Security settings' the Key Content field reveals the stored Wi-Fi password in plaintext.

```

C:\Windows\System32>netsh wlan show profile name="Asd" key=clear

Profile Asd on interface Wi-Fi:
=====

Applied: All User Profile

Profile information
-----
Version                : 1
Type                   : Wireless LAN
Name                   : Asd
Control options        :
    Connection mode    : Connect automatically
    Network broadcast  : Connect only if this network is broadcasting
    AutoSwitch         : Do not switch to other networks
    MAC Randomization  : Disabled

Connectivity settings
-----
Number of SSIDs        : 1
SSID name              : "Asd"
Network type           : Infrastructure
Radio type             : [ Any Radio Type ]
Vendor extension       : Not present

Security settings
-----
Authentication         : WPA3-Personal
Cipher                 : GCMP-256
Authentication         : WPA3-Personal
Cipher                 : GCMP
Authentication         : WPA3-Personal
Cipher                 : CCMP
Security key           : Present
Key Content            : 12345678

Cost settings
-----
Cost                   : Unrestricted
Congested              : No
Approaching Data Limit : No
Over Data Limit        : No
Roaming                : No
Cost Source            : Default

```

3. Conclusion of the wireless part: any user with administrative access to a machine can extract every saved Wi-Fi password, showing why physical access to a system is itself a security risk.

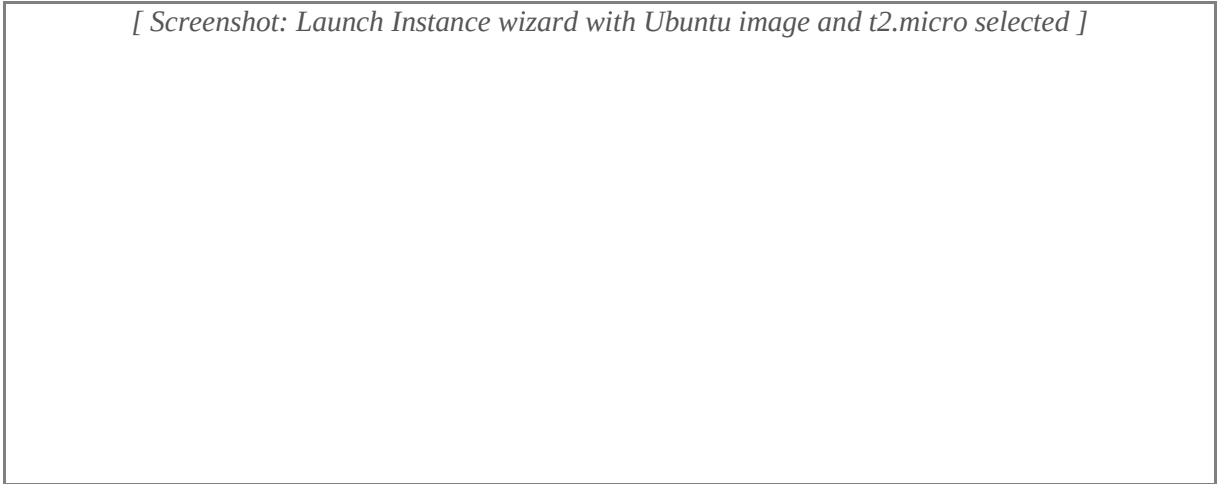
4. For the cloud computing part, open <https://aws.amazon.com> (or <https://cloud.google.com>) and create a free-tier account. Login to the management console.

[Screenshot: AWS Management Console / GCP Console homepage after login]



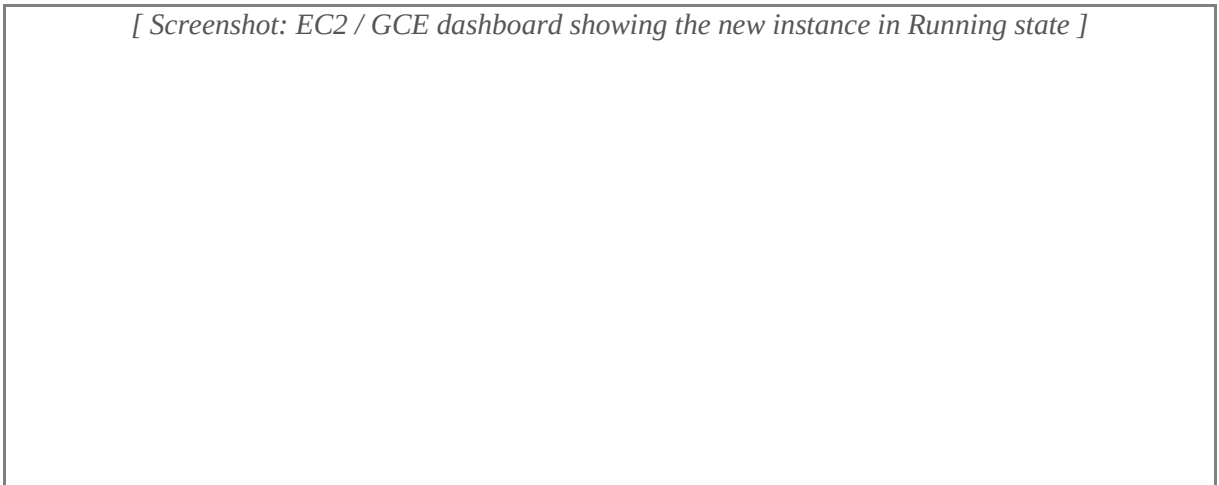
5. Open the EC2 service (or Compute Engine on GCP) and click 'Launch Instance'. Choose the Amazon Linux 2 / Ubuntu free tier image and the t2.micro / e2-micro instance type.

[Screenshot: Launch Instance wizard with Ubuntu image and t2.micro selected]



6. Generate a new key pair, allow SSH (port 22) in the security group and click 'Launch'. Wait until the instance status is 'Running'.

[Screenshot: EC2 / GCE dashboard showing the new instance in Running state]



RESULT: Thus, saved Wi-Fi credentials were recovered using the netsh wlan command, and a free-tier virtual machine was launched on a cloud platform successfully.